Unit 6: File management and storage structure
File organisation
concept of file and directories
system case for the file system.
space management.
Data structures INODE, superblock
and virtual file system.
Disk layout and formating.

Book: OS concept by Gilber charts and galvin

Machine are
stable, predictable, structured.

compound, speed, I/O (Language).

Software:
~~To bridge the gap betⁿ application domain~~
The semantic gap betⁿ the application domain (human) and the execution domain (machine)

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is breiged by a logical entity termed as software.

Software is the group of progⁿ that helps in bridging the semantic gap betⁿ AD and ED.

Different types of software.

System Software — Application software.

→ Special system software → compiler, Assembler. (to translate)
→ Software tools → editor, Debuger etc.
→ Operating system.

Application software: Railway, Banking

☆ Operating system is a system software that takes the care of operation, execution and maintainance of the valuable resources.

System software:
System software are the programmes which aids in effective execution of general users computational requirements in the computer system.

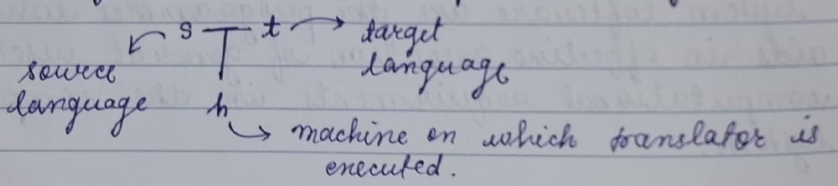
It is a collection of programmes which are needed in creation, preperation, execution of the users programme.
They are the programs which utilize the resoures of the system in optimised way and

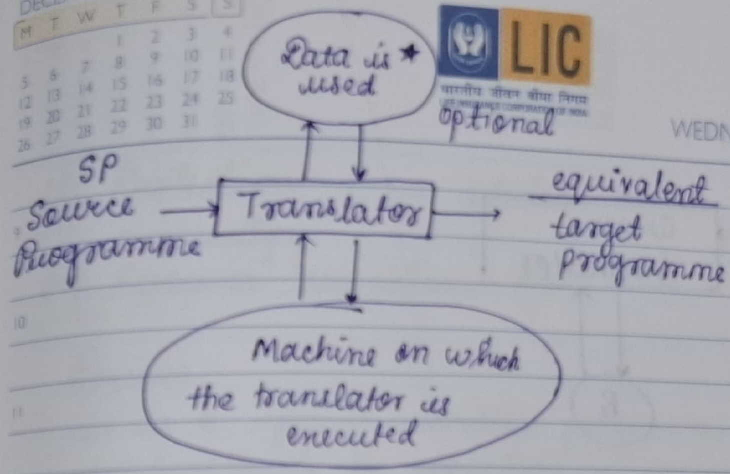
helps the user to translate the programmes and execute it.

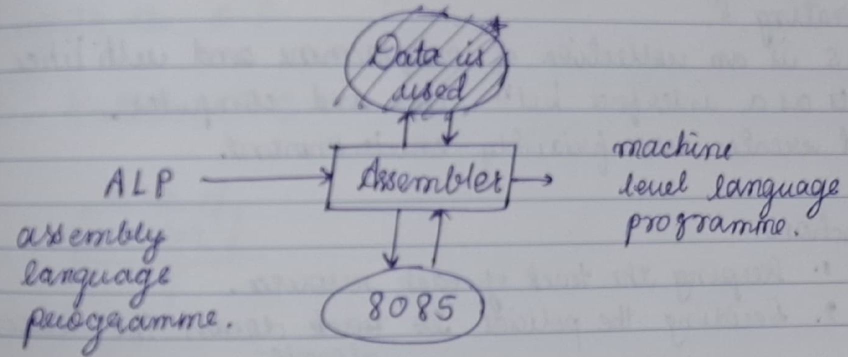
Translators

compiler → complete programme translation
interpreter → line by line
assembler.
linker
loader
cross compiler
stimulator
Debugger
Preprocessor
Macroprocessor
Discompiler or/Disassembler.

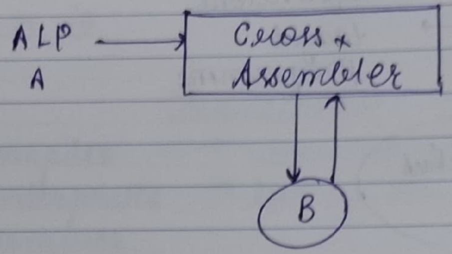
For any translator is represented by 'T'. and







The information regarding the address entries is placed in the list.



Operating S
O.S is an collection of programme and utilities acts as a interface betⁿ user and computer. and creates user friendly environment.

functions:
1. Keeping the track of each resource.
2. Deciding the policies to ~~track~~ allocate down the resources.
3. Delocate the resource.
4. Physically allocate the resources.

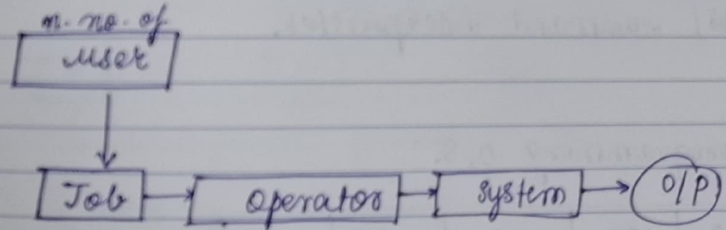
Diff. services offered by operating system.

components.
(i) Programme execution
(ii) I/O operation
(iii) File system incrementation
(iv) communication (communication betⁿ Processes.)

(v) Protection: error detection.
(vi) accounting
(vii) Resource allocation.

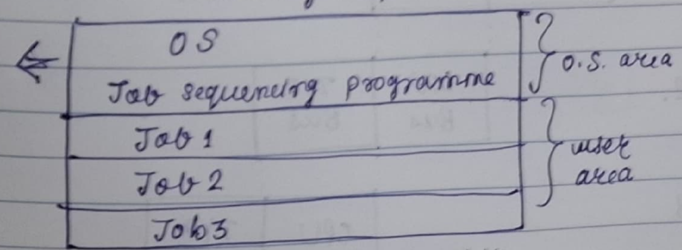
Driver:
it is a special written programme which understands the operation of device and interfaces with the O.S.

1. Batch O.S.



it is the first operating system. in this O.S. the user has to submit programmes to the operator. then operator will put it for the execution.

memory map.



M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Disadvantag:

1.

CPU Bus	I/O Bus

← Job →

Programme

CPU → CPU bounded

I/O device. → I/O bounded.

~~2. Multiprogramming O.S~~

1) main O.S system ← O.S.
2) Residential Programme (TSR)
3) Device Drivers Terminal / 8
4) control / command interpretter.

2. multiprogramming O.S.

CPU bus	I/OP Bus	CPU Bus	I/OP Bus	CPU Bus	I/OP Bus.
Job 1		Job 2		Job 3.	

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Job 1

CPU bus	I/O Bus

Job 2.

CPU Bus	I/O Bus

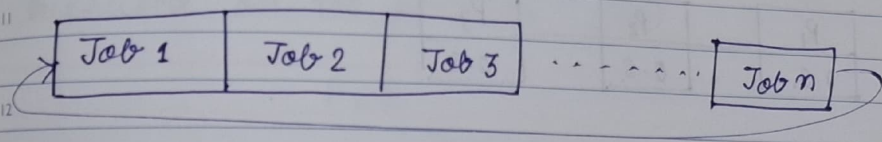
Job 3

CPU	I/O Bus.

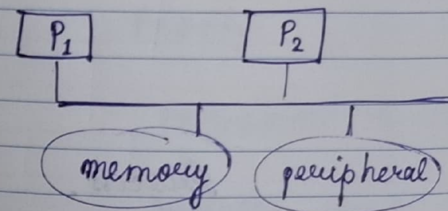
M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Differentiate multiprogramming and multitasking.

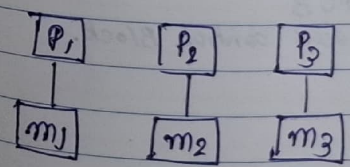
3. Time Sharing. O.S.



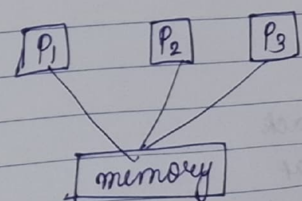
Multiprocessor. O.S.
→ network O.S.
→ Distributed O.S.



loosely coupled

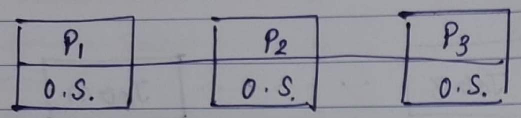


tightly coupled.

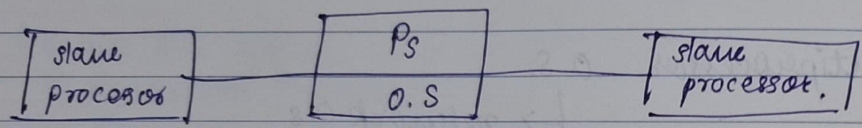


1. Separate superviser multiprocessing system.
2. Master slave MPOS system.

1. each has separate copy of O.S.



2.



⇒ Process management.
↓
* CPU management.

Programme — Process.

→ secondary memory. → main memory.
→ binary code → code + data structure
+ PCB
process control Block.

Stack
heap
variable
a out

Process is the fundamental task of O.S. it is a programme execution or an instance of a programme running on a machine or it is entity that can be assigned to and executed on the processor (CPU).

PCB

Identifier

State

Priority

Programme counter.

memory pointer

content data.

I/O status

Advanced O.S.

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Identifier is a unique id created by the process for the programme to be executed.

~~State~~ every proces has to passed throug diff. states through its life cycle. The status is sho

3. Priority:
diff programes has diff priority assigned by O.S.

4. Programme counter:
in the programme counter the address of next instruction is stored.

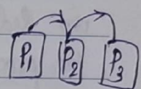
5. Memory pointer:
it is a starting address of process to assign it also includes the memory paint for programme code and data code.

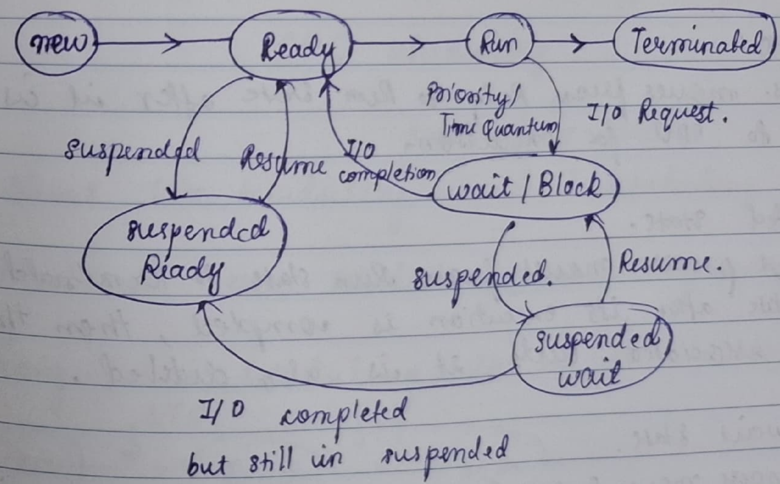
6. Content data:
when programme is executing it uses seeme registors that contains an information i.e- a content data.

7. I/O status information:
when ever a prosses is req. in I/O. and if I/O is asigned to that press. then its status is mentiond in I/O. status block.

8. Accounting infor:
this may include the amount of processor time clock time used time limits etc.

→ Linked list is used for the programme control block.





new State:
A process is said to be in new state when the progname pressent in secondary memory and initiated for execution.

Ready state:
A process moves from new state to Redy state after it is loaded into the main memory and it is ready for execution.
in Redy state the process waits for the execution by the processor.
in multiprogramming enviroment many processes are in the redy state.

Run state:
A process. moves from Redy to Run state after it is assigned to CPU for execution.

Terminated state.
A process moves from Run state to terminated

state after its execution is compled, then the PCB associated with it is also deteled.

Block/wait state.
A process moves from Run state to block state by 3 Reason.

1- Request I/O
2 High priority press is assigned for execution
3- Time quantum is finised in time sharing O.S

Suspended Ready.
A process moves from Ready to syspended Ready if the process from ↑ priority is to excuted but main memory is full.

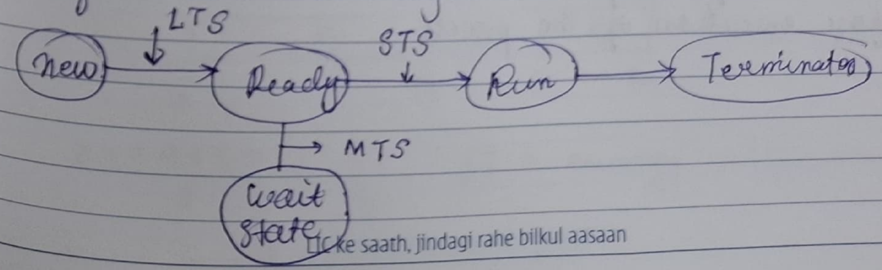
Suspended wait:
A process moves from wait state to syspended wait if the process with ↑ prio is to be executed with som I/O devices but main memory is full.

CPC scheduling:
↓
scheduling Policies.

Short Term scheduling : CPU scheduling.

Medium Term scheduling: memory swaping

Long Term scheduling: to see all situations.



CPU scheduling.

Short Term Scheduling :

shortest Job first SJF — primitive — non-primitive → FCFS.

First come First Serve.

- arrival time.
- waiting time.
- Bus. time.
- completion time.
- Response time
- Turn around time:

When a process enteres into the Redy que that time is called Arrival time.

Waiting time is the amount of time spent by a process waiting in the redy queue from getting the CPU. ~~time~~ for execution.

Burst time is the amount of time required by the process for execution on CPU.

Bus time of the process can't be known in advance before execution of the process.

Completion is the point of time at which process completes its execution and terminated from CPU. also called exit time.

Response time is the time when the process gets the CPU for the first time after entering into ready queue

Total amount of time spend by a process in the system.

TAT = Comp time − Terminating Time.

Q. consider the foll. processes with gear CPU bus time. Calculate average waiting time.

	Processes	Bus Time.
$P_1 - 0$	P_1	24s
$P_2 - 24$	P_2	3s
$P_3 - 27$	P_3	3s

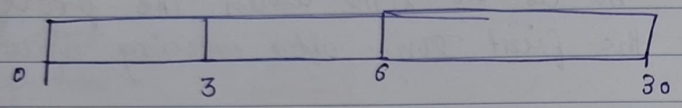
→ Gantt Chart.

FCFS

P_1	P_2	P_3
0 – 24	24 – 27	27 – 30

$$\frac{0+24+27}{3} = \frac{51}{3} = 17$$ = average waiting time.

	Process	Bust time.
$P_1 - 0$	P_1	3
$P_2 = 3$	P_2	3
$P_3 = 6$	P_3	24



$$\text{average} = \frac{0+3+6}{3}$$

$$= 9/3$$

$$= 3$$

→ Convoy ~~Convoy~~ effect
∴ average time is incrased.

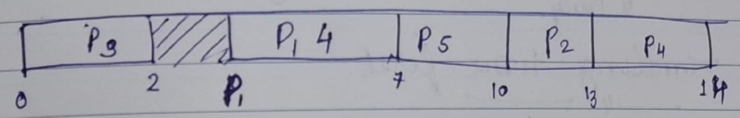
~~Cor~~

Q. Consider the set of 5 process whose arival time and bus time is given below. Calculate average waiting time

Process	Arival Time	Bus time.
P_1	3	4
P_2	5	3
P_3	0	2
P_4	5	1
P_5	4	3.

FCFS

P_3	0	2
P_1	3	4
P_5	4	3
P ~~2~~	5	3
P ~~5~~ 4	5	1



arival time.

Calulate Turn around time, finish - ~~start time~~

NOVEMBER 2022

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Premitive
SJF (shortest Job First.)

LJF → priority
Round Robin → (time sharing)
Policy.

Short remaing time first
Longest — " —

highest response ratio next.
multiple queue schedulding.
miltilevele queue scheduling.

	arival time	bus time.

Q. consider the situation for following processes.

P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5.

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Find average waiting time by the policy shortest job first with premision and without primsion.

DECEMBER 2022

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Ans: non-primitive.

P_1	P_2	P_4	P_3
0 — 8 unit — 8	4 unit — 12	5 unit — 17	9 unit — 26

waiting time =
~~$P_1 \to 0$, $P_2 \to 8-1$, P_4 $5-4$, P_3 $9-7$~~
$= \cancel{0 + 7 + 1 + 2}$
$= \cancel{10}$

$= P_1 \to 0,\ P_2 \to 8-1,\ P_3 = 17-2$
$P_4 \to 12-3$

$w.T = \underline{0 + 7 + 15} + 9$
$= \boxed{31}$

$avg.\ w.T = \frac{w.T}{4}$

Primitive:

P_1	P_2	P_4	P_1	P_3
0 — 1	5	10	17	26

$w.T = P_1 (10-1=9) + P_2 (\overset{1-1}{0}) + P_3 (\overset{17-2}{15}) + P_4 (\overset{5-3}{2})$
$= 9 + 0 + 15 + 2$
$= 11$
$= \cancel{\frac{15}{26}}$

H.W.

	Arriving.	Bus time.
P_1	0	7
P_2	2	4
P_3	4	1
P_4	5	4

N.P.
avg w.T = 16

P
avg w.T = 3.

Round Robin Policy:

Time unit. = 20

	Bus		
P_1	53	33	13
P_2	17	0	
P_3	68	48	28
P_4	24	4	

$$W.T_{avg} = \frac{(\quad)}{4}$$

first come first serve.

57
20

Ans:

P_1	P_2	P_3	P_4	P_1	P_3	P_4
0 – 20	20 – 37	37 – 57	57 – 77	77 – 97	97 – 117	117 – 121

$20 \leftrightarrow 37$
17

P_1	P_3	P_3
– 134	134 – 154	154 – 162

134

avg w.T = $P_1 \rightarrow 0 + (77-20) + (121-97)$ ~~+ 134 −~~

Q. Consider

P_1	30	unit
P_2	6	unit
P_3	8	unit

quantum time = 5 unit.

avg w.T = ? = 15 unit.

~~Dam~~ Dham
~~Dha~~ Rt. Dhere

Priority Policy.

	Prio	Bus.	Arrival
P_1	10	3	3
P_2	1	1	4
P_3	2	3	0
P_4	1	4	1
P_5	5	2	1
	Bus,	Priority	

avg W.T = 08.2

P_2	P_5	P_1	P_3	P_4
0 –	– 6	6 – 16	16 – 18	18 – 19

P_2